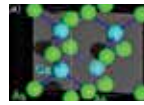


Teaching human cells to delay Aging

Key process in all human cells that contributes to diseases like cancer and neurodegenerative diseases as well as aging. <https://digit.in/feb19-81>



5000 times faster than Computers

Using an Interatomic light rectifier researchers have generated directed currents at terahertz frequencies. <https://digit.in/feb19-82>

SHRINK SCIENCE

Researchers have developed affordable nanoscale shrinking tech, but you still won't get an Ant-Man suit. Here's why.

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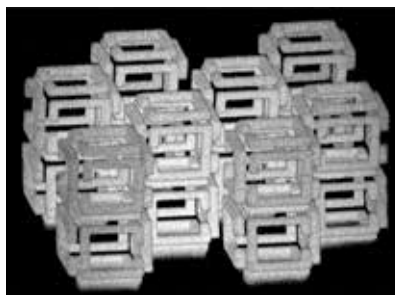
It's been a while since we've seen shrinking technologies in movies. Shrink objects and even

people in some instances makes for great cinema. The latter constitutes the more popular instances such as Honey I Shrunk The Kids, Fantastic Voyage, and the more recent Ant-Man movies. In most of these cases, there's some fictional explanation for the science behind all of it. For instance, in Ant-Man's case, Dr Hank Pym discovers 'Pym particles' which allow him to develop technology that can shrink or enlarge things almost indefinitely. While it doesn't need to be explained that it's pretty much impossible to have that level of control over your mass and volume, as it turns out, shrinking tech isn't as fictional as you might think.

SHRINK FAB

No, we haven't invented the Ant-Man suit or shrink rays yet. However, Massachusetts Institute of Technology (MIT) researchers might have brought us as close to both as we possibly can be. The researchers have invented a way to fabricate nanoscale

3D objects of almost any shape, a process that involves replicating an object on much smaller scales. This object creation method can be augmented further since the researchers can also pattern the objects with materials such as metals, quantum dots and



This image shows a complex structure prior to shrinking

DNA. This is a fabrication process at the nanoscale in three dimensions using any kind of material.

In this new technique, researchers can create a nano-scale version of any object they want by patterning a polymer scaffold with a laser. Once that is settled, more usual or useful material is added to the scaffold before shrinking it to nearly 1000th of the original. This process is known as Implosion fabrication and it uses materials that are found possibly in every laboratory around the world, and even in children's diapers.

IMPLOSIVE

The need to develop this method comes from limitations of existing nanoscale fabrication technology. For instance, the method that uses etching structures onto a surface using light is good enough for 2D structures but obviously fails with 3D structures. For 3D nanostructures, researchers have to add layers upon layers of material in a slow and painstaking process. This process can also only result in self-supporting structures, such as a pyramid but not like a hollow sphere. Additionally, the materials that can be used for such a method are limited to polymers and plastics, which do not possess the characteristics necessary for quite a lot of applications.

Edward Boyden, the Y. Eva Tan Professor in Neurotechnology and an associate professor of biological engineering and of brain and cognitive sciences at MIT, along with his team decided to overcome these limitations using a method that was developed by his lab a few years ago. The technique is known as expansion microscopy and was used to develop high-resolution imaging of brain tissue. This technique involved embedding tissue into a hydrogel and expanding it before subjecting it to a regular microscope. Boyden and his students decided to reverse the process.

What that means is - they embedded larger objects into already expanded hydrogel before shrinking it to the nanoscale, aka implosion fabrication. Highly absorbent polyacrylate, commonly found in diapers, was used